# ICT30016 IoT Track – Week 1

## The Project Proposal & Architecture Design

**Prepared by Dr. Mark Tee Kit Tsun (SUTS)**

# An IoT (or general Software) Proposal




- There is no single standard Form to fill (unless the client has one).

- However, every Project Proposal should have these components:
  a) **Background / Problem Description**
     - Describes the client company, operating environment, what problems are faced, and why the clients need a solution.
     - Written from your (the Development Team's) point of view.
     - Shows your understanding of the Client's issues.

  b) **Scope Statement**
     - What is the aim and capability of the proposed solution.
     - Why can it solve the client's problem(s).
     - Usually consist of:
       - Goals – the general purpose of the solution.
       - Objectives – specific attributes used to measure the success of the project (must be S.M.A.R.T.).
       - Constraints – Conditions and Limits that must be adhered to.
       - Out of Scope – Specific things that the Team will not be working on.

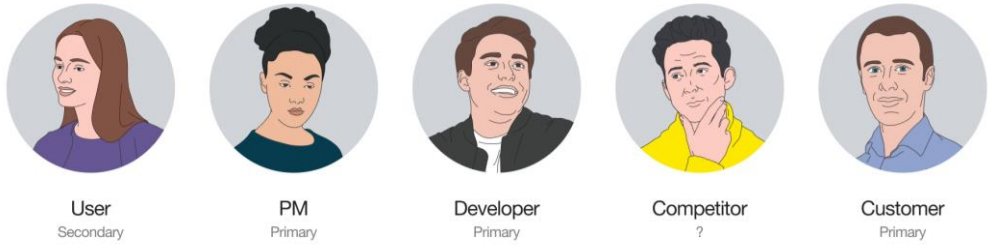



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# An IoT (or general Software) Proposal

**c) Stakeholders**

- Who is involved in the project.
- Can include Project Sponsor, Customer/Client, Developers, and/or more.
- Describes the Users who interact with the system.




**d) Deliverables and Schedule**

- Indicate what are the items to deliver, and when (e.g., Installers, Servers, Nodes, Manual, etc.)
- The Timeline:
  - use calendar dates and Sprint Numbers to indicate when the expected completion and submission of deliverables is.
  - Use a Gantt Chart to visualize it.
  - Consider things like Dependencies, Risks involved and Business Value.




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# An IoT (or general Software) Proposal

**e) Solution Direction**

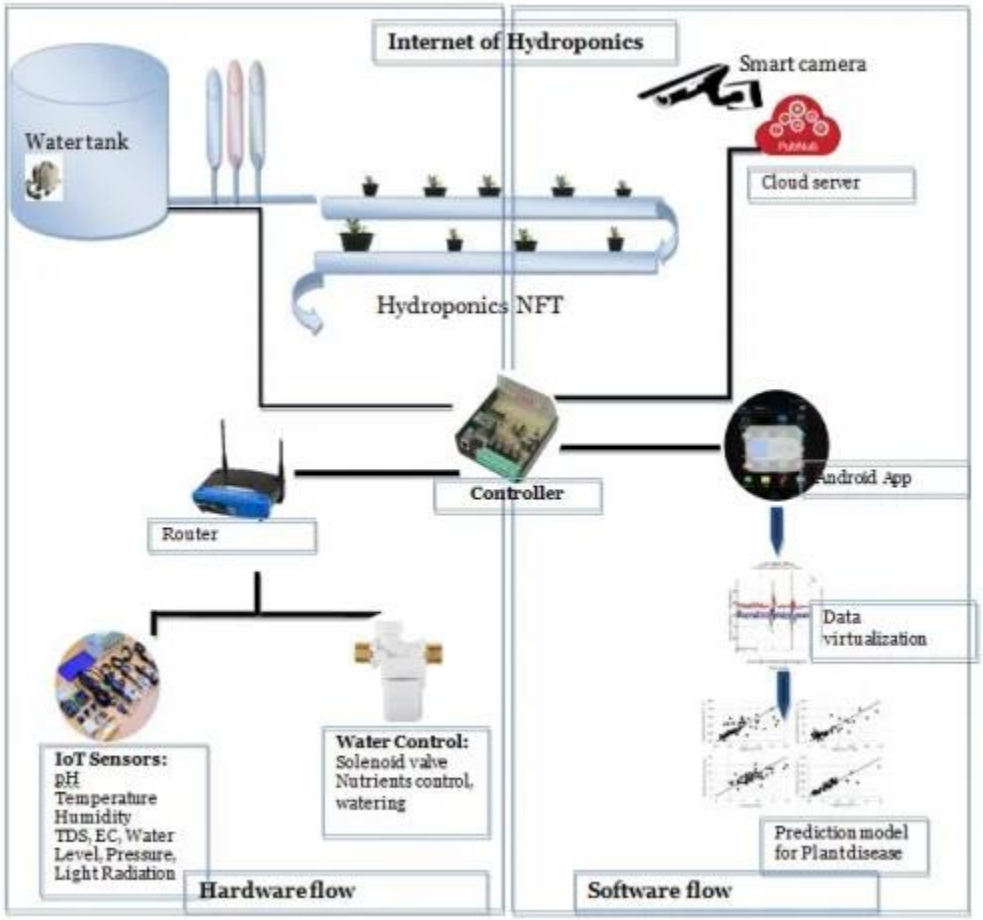
- High-Level Description of the overall system
- How will the system work
- What technologies are involved in building and operating the system
- Provide a baseline IoT and Software Architecture

**f) Quality Management**

- Define the characteristics of Quality that this project must meet.
- This definition of “Done” is used to check whether a developed item is qualified enough to meet the requirements of the client.
- ***Quality Assurance*** is the management of this process throughout the project.

**g) Resources**

- This section of the Proposal describes the materials, manpower and equipment gathered for the project.
- Can be simply divided into ***Human Resources*** and ***Non-Human Resources***.




Schematics of Internet of Hydroponics

# Case Study: IoT-based Accident Detection & Response

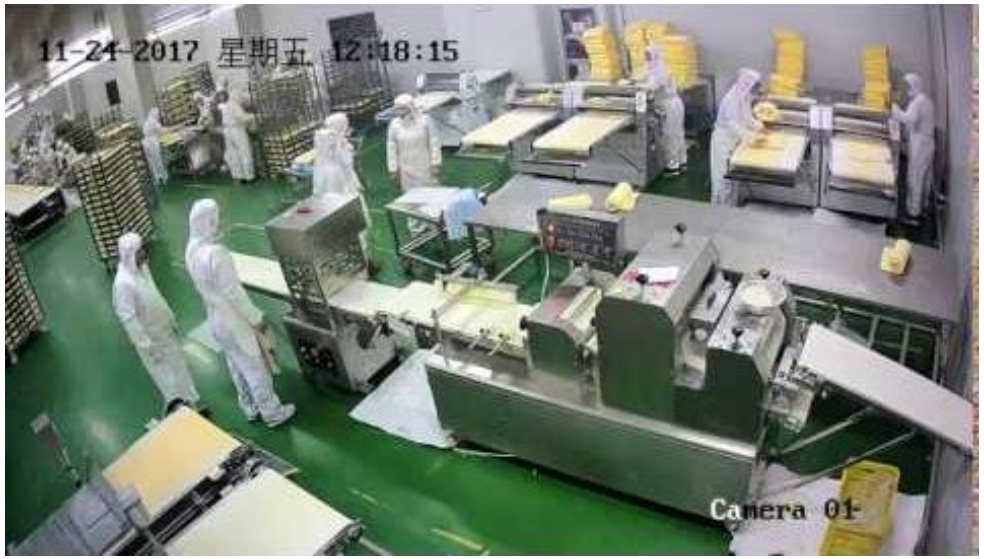
**Situation:**
Your team is tasked to meet an industry client who owns a factory. This facility has a long history of frequent accidents because it is a fast-paced working environment full of employees moving around heavy machinery.

All employees are only tracked when they punch into their shift. Afterwards, they proceed to all areas of the factory workspace, which spans a very wide area. An accident is only reported by employees who witness the occurrence and activate a general Alarm. Most injuries become fatal because emergency staff take too long to find out where the accident is, and which types injuries are expected.

They need a solution to this problem which does not interfere with the factory's physical workspace, not require additional infrastructure other than its present Wireless LAN and is informative to non-ICT personnel in the Occupational Safety and Hazard planning team.






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# Case Study Step 1: Problem Description

**Identify the issues/problems:**

- Employees are only tracked when the enter or leave the building (through punch card). They can only be identified when their Employee Card is read by the emergency staff.
- Accidents are only indicated by the general alarm triggered by Witnesses.
- The type of injuries can only be addressed when the Emergency Staff reach the location of the Accident and assess the victim/s.



**Pain Points:**

- Too much time passes between the occurrence of the Accident and Emergency Staff reaching the Victim, increasing the rate of fatality.
- Emergency Staff must guess where to go, based on the general direction of the Alarm.
- Emergency Staff must guess what injuries to prepare for, based on general direction of the Alarm.





**What Issues should be clarified on?**

**Rough Solution:**

An IoT system that monitor every employee and notifies Emergency Staff in case of a possible injury, based on who was affected, and in which part of the factory.

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# Case Study Step 2: Scope

- We need a clear set of Scope Description (what does it do) and Objectives (what are the general, realistic goals).

- Scope must be a short description of the primary purpose/functions of the system to respond to the Problem Description. Example:
  - *This IoT system will involve the development of an activity tracker for each employee to track their general motion and orientation patterns.*

  - *In case an aggressive pattern is detected (indicating possible accident), it will notify the platform via WiFi and the Internet, along with the employee's identity and the identity of the connected WiFi Access Point (which will help identify the location of the accident).*

  - *The notification will inform Emergency Staff of who is involved, where the accident is located and what injuries to prepare for, based on the type of machinery at that location.*

  - *By collecting a history of accident occurrences, analytics can help identify hotspots to focus prevention measures.*

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# Case Study Step 2: Scope (Cont.)

- The Objectives are attributes or general goals of the project, which addresses the Pain Points and Constraints.
- Must be S.M.A.R.T.
- Example:
  - *Near real-time notification of accidents upon detecting possible pattern indicating an accident.*
  - *Indication of accident location within 30 meters radius of an alarm station.*
  - *Advice on injuries expectations consider the machines at accident location.*
  - *Implementation consume none of the ground workspace and minimize obstruction of employee work.*
  - *Rely on existing network to facilitate communications.*

*From here onwards, determining **Step 3: Stakeholders** then setting the **Step 4:** Schedule (**Timeline**) and **Deliverables** should be straight forward.*

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# Case Study Step 5: Solution Direction?

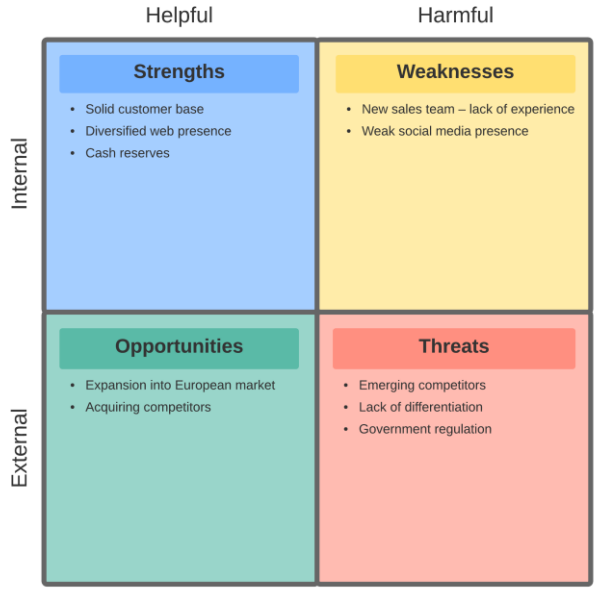


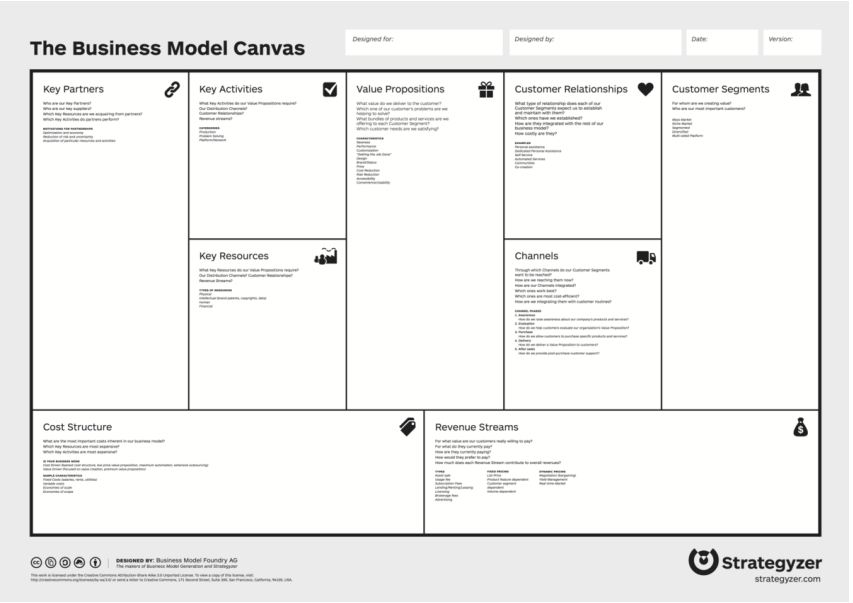

- Derive Potential Solutions. Here are some pointers:
  - Identifying the Problem Domain – tap into knowledge of this domain
  - Choose technologies commonly used in this domain
    - Avoid technology gaps
    - Make use of existing and tested tools/techniques for these kinds of projects.
  - Use the ***Project Forces*** to describe the Solution Direction
  - Visualize your Potential Solutions using High-Level Designs (block diagrams and flowcharts are useful).
  - Try to think outside the box, but not outside the constraints!
  - Evaluate the **Feasibility** of each Potential Solution, according to the Constraints

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# Case Study Step 5: Solution Direction (Cont.)

- *Now, choose a Solution:*
  - *Apply the Objectives as a criteria to select one from the pool of Potential Solutions.*
  - *Perform Risk Analysis – to identify the threats to the project*
    - *One example way is KoST – identify Knowledge, Skills, and Technology Gaps*
  - *Cost-Benefit Analysis – which is the best for the client?*
  - *Feasibility Analysis – is this solution possible and realistic?*
  - *Some projects involve defining a **Business Case** for the chosen solution (e.g., Business Model Canvas)*
- If you cannot decide, that means you have not gotten enough information yet!



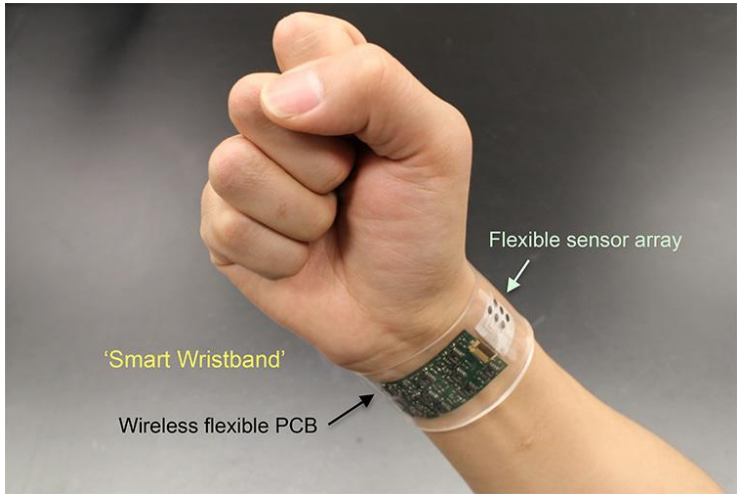




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# Case Study Step 5: Solution Direction (Cont.)

- Now back to the Case Study! Example:
  - Problem Domain: Activity Tracking
  - Potential Solutions:
    - Vision-Based Activity Tracking using Kinect – Rejected! Involves heavy installation in the workspace.
    - Mobile App – Rejected! From Client interview, it appears that Employees are not allowed to bring personal devices into the facility.
    - IoT Machine Monitoring System – Plausible! Installation of vibration sensors and status monitoring of machines into remotely connected nodes. Uses AI to learn the common operation patterns of machines so anomalies indicate accidents. But Rejected! due to wide Technology Gap (need to incorporate advanced Machine Learning) and short development Time.
    - Smart Wearable Activity Tracker – Accepted! The implementation is non-obstructive as it is worn as a watch by employees.



# Risk Drivers using KoST Gap Analysis

- In any project, there is the potential for difficulty caused by lack of knowledge, skills and appropriate tools
- It is important at the outset of a project to consider this
- One approach is called “KoST” analysis
- Note : This is not “famous”, and you will not easily find references to it! But it is a useful first look at project risks – one which, if not carried out, could lead to disaster!
- (In the Myki project in Victoria, of the 6 tendering providers, 5 had the experience of a successful smart-ticketing system implementation; the contract was awarded to the sixth provider! Enough said!!)

Principal risk drivers (KoST):

- Knowledge Gap (*don't know*)
- Skill Gap (inexperience)
- Technology Gap (unknown/young or unavailable)

*KoST analysis* attempts to determine

- knowledge and skill gaps a team/individual has, and
- whether any technology gaps exist.

Note that in general there will be many other risks associated with a project, and KoST is not a full risk analysis – covered in a later lecture.

- Knowledge Gap
  - Lack of knowledge of the problem domain, and the particular context of the desired solution
- Skill Gap
  - Lack of the skills needed to implement the system according to the proposed approach
- Technology Gap
  - Using a technology that is new, without certainty that it will work for the proposed solution {more than just a skill gap with the technology, but also an uncertainty about whether the technology can actually work in this context}

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# Architecture Design

- **Systems Modelling** is a process of conceptually visualizing the ***structure*** and ***behavior*** of a system.
- There are various methodologies to do System Modelling. In Software, the most prominent way is the **Object-Oriented Approach**.
- The output of this process is a set of documentation that describe the proposed system's ***structure*** and ***behavior***, called Architecture Design.
- There are 2-Levels of Architecture Design:
  - High-Level Design
    - Shows the Overall Structure but not the exact mechanism in each module.
    - Translated from the main Use-Cases / Usage Scenarios.
    - Also based on meeting non-functional requirements (e.g., reliability, throughput, consistency, etc.).
    - Normally do not change.
  - Low-Level Design
    - Shows the detailed mechanism and components structure in each module.
    - Must account for Programming Language, Platform, Technology and Hardware.
    - Must be detailed enough to be implemented using Programming Language, Platform, Technology and Hardware.

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# High-Level Architecture

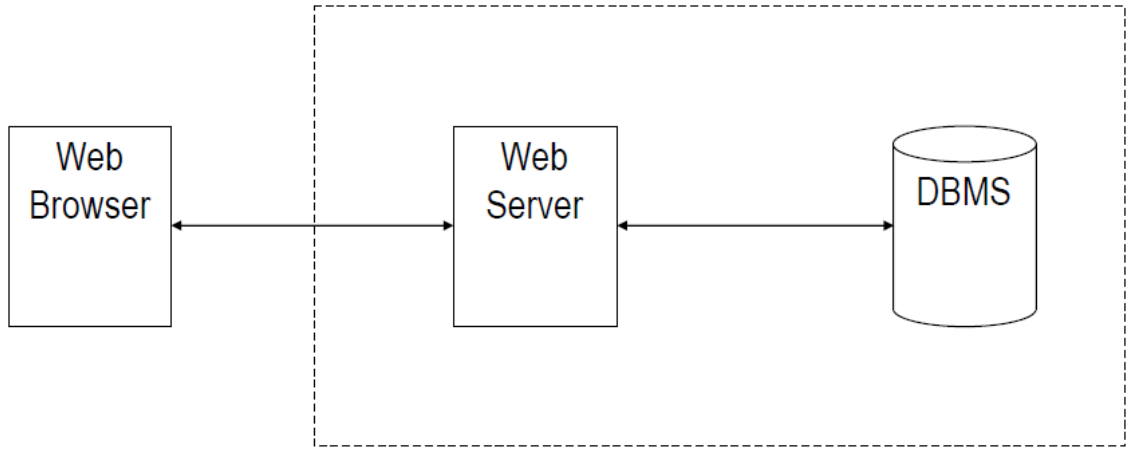
- This must be present in the **Project Proposal** to inform the Client that this is what the proposed solution will look like.

- It Captures the gross structure of the system:
  - How it **consists** of software and hardware components
  - How the **interactions** between components take place
  - What are the key **properties** of the components

- Why is this important?
  - Handling **Complexity**
  - **Communication** among Stakeholders
  - Early **Design Decisions**
  - Architecture is **Transferable** and **Reusable**

- A good Architecture Design:
  - Uses consistent set of principles and techniques
  - Resilient to changes
  - Source of guidance across the product timeline
  - Reusable

■ Example: Web application



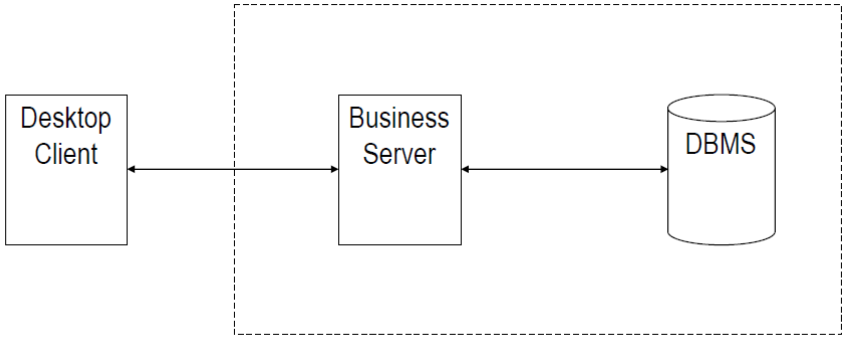

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# High-Level Architecture (Simple Software Examples)

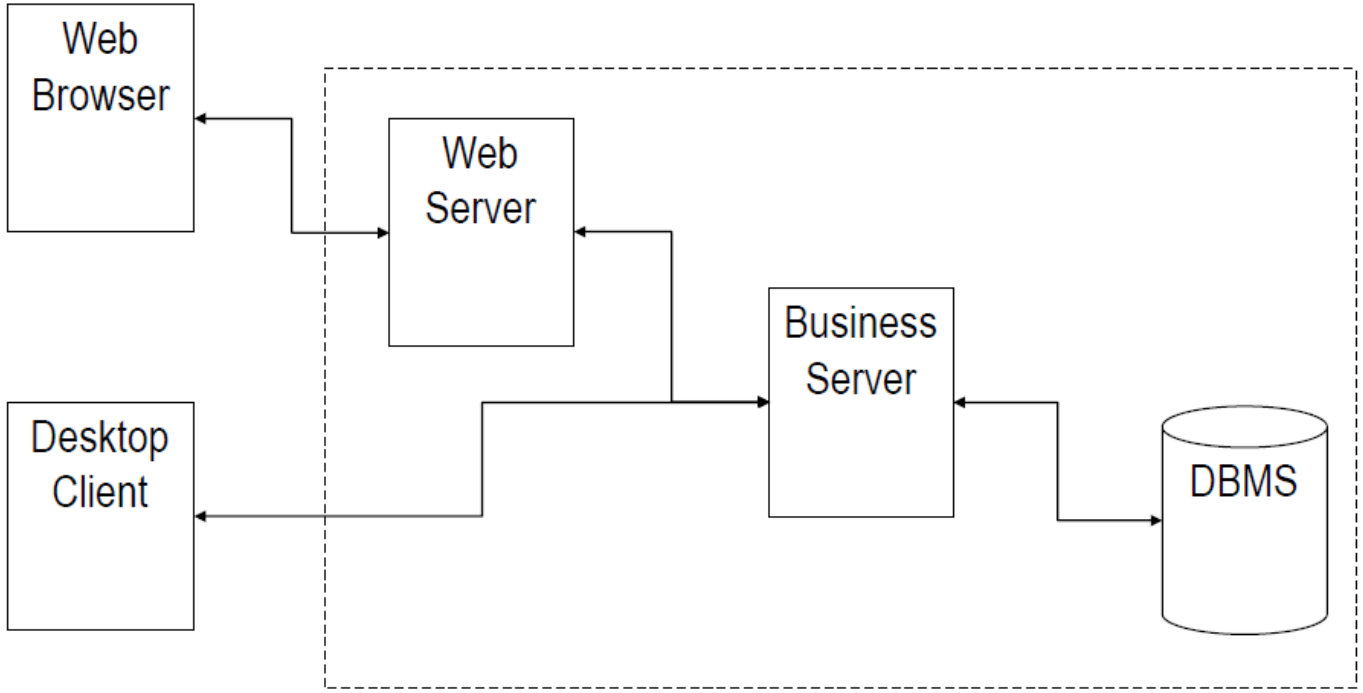
- Example: Desktop application (old days)



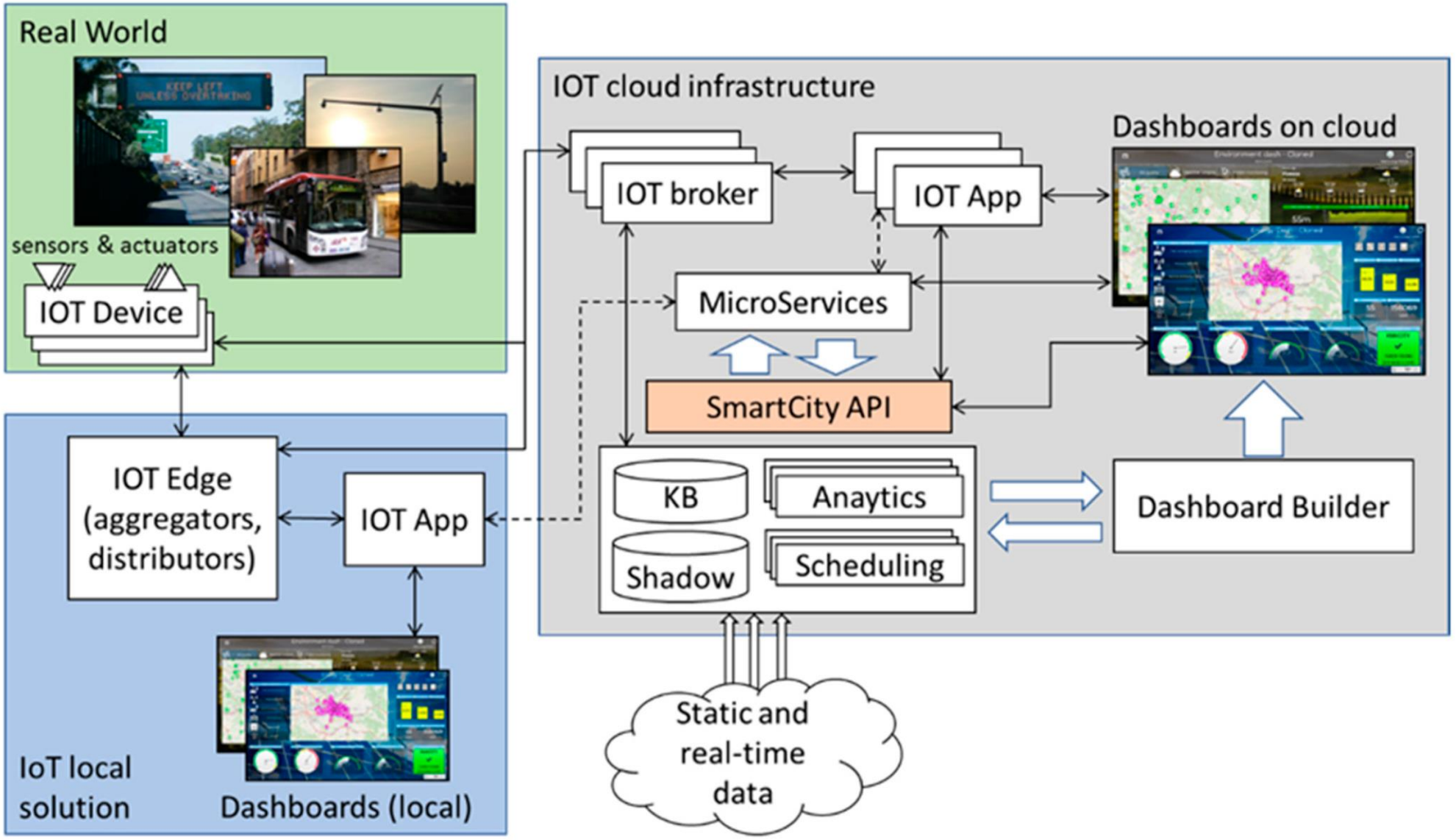

- Example: Desktop application



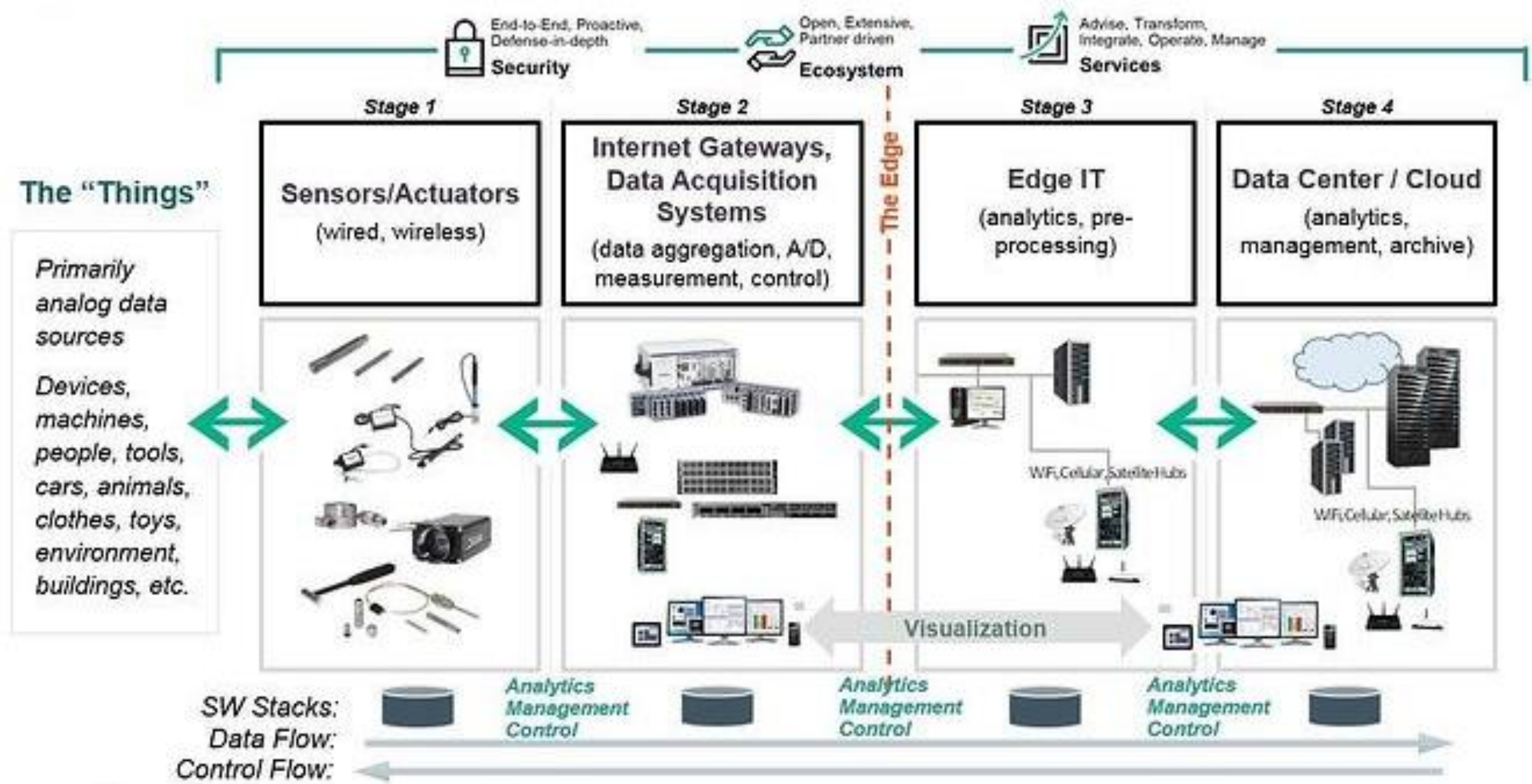

## Enterprise Framework



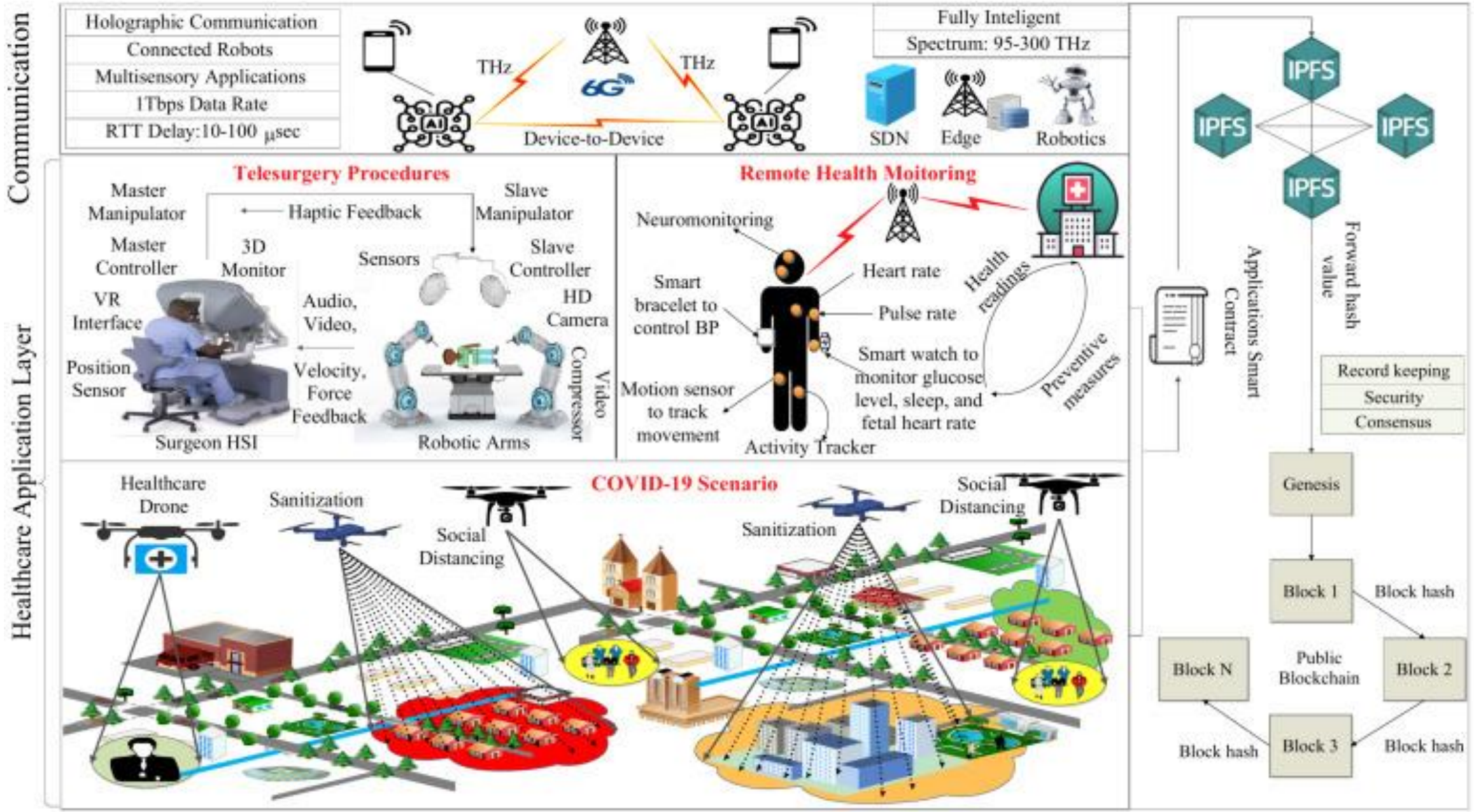
# High-Level Architecture (IoT Examples)



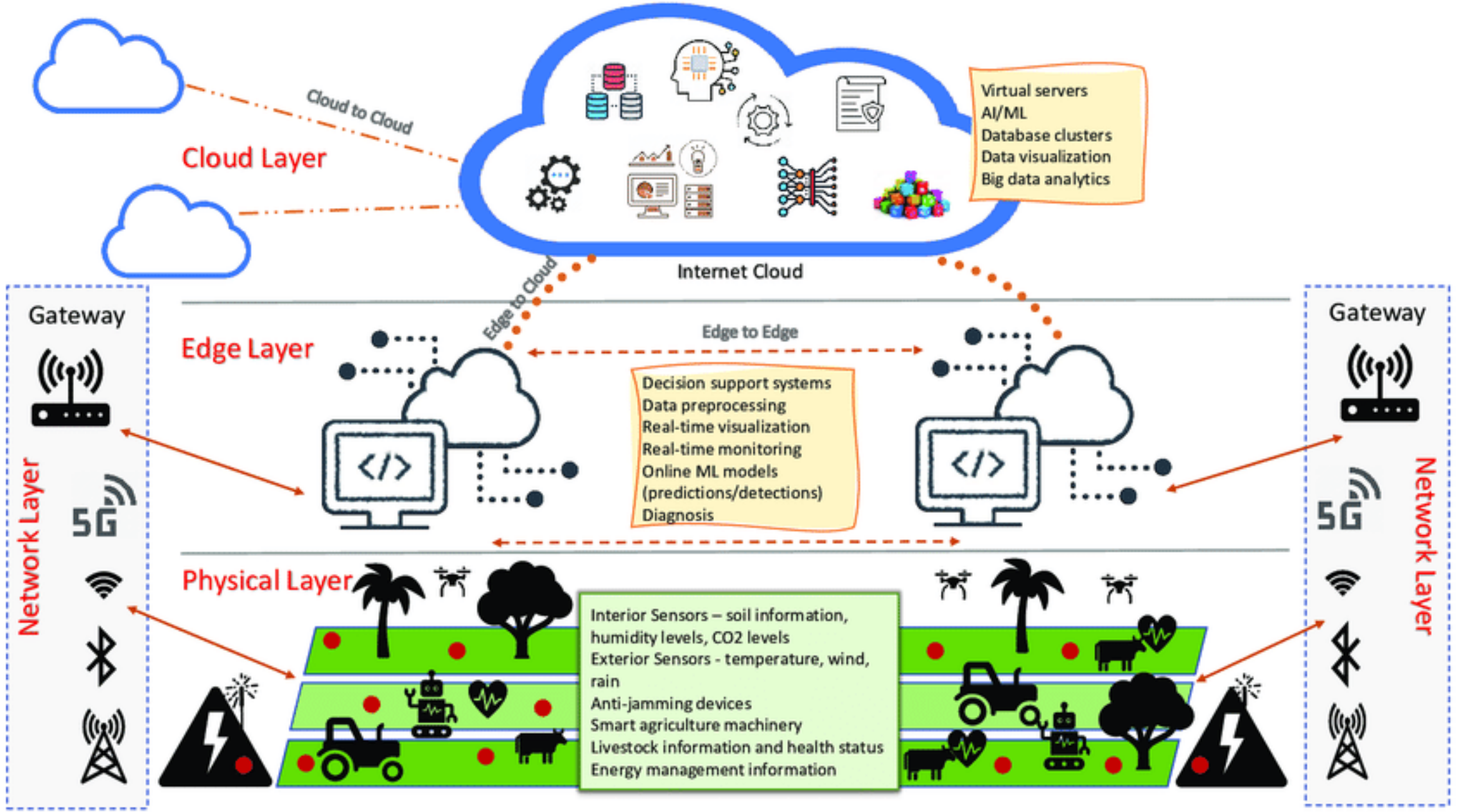
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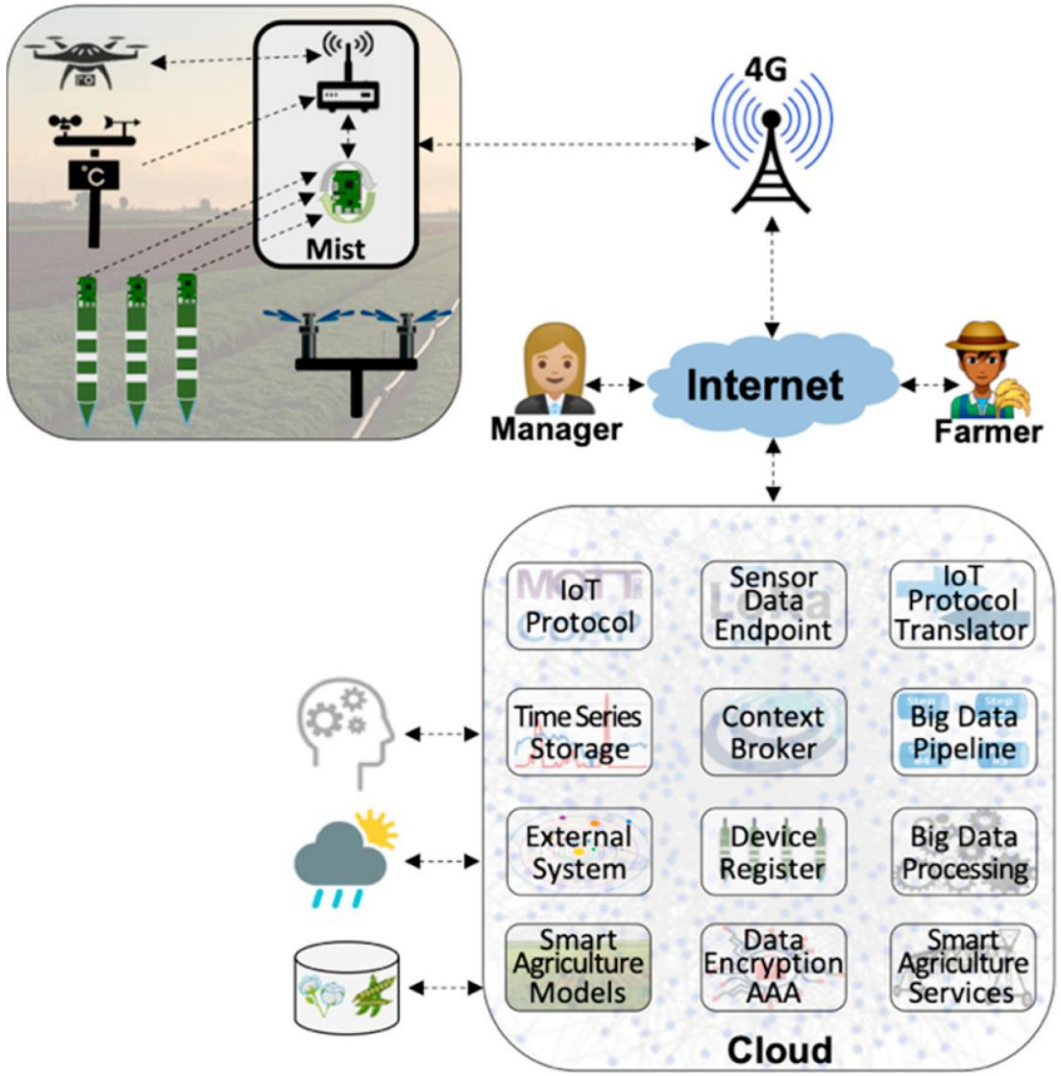
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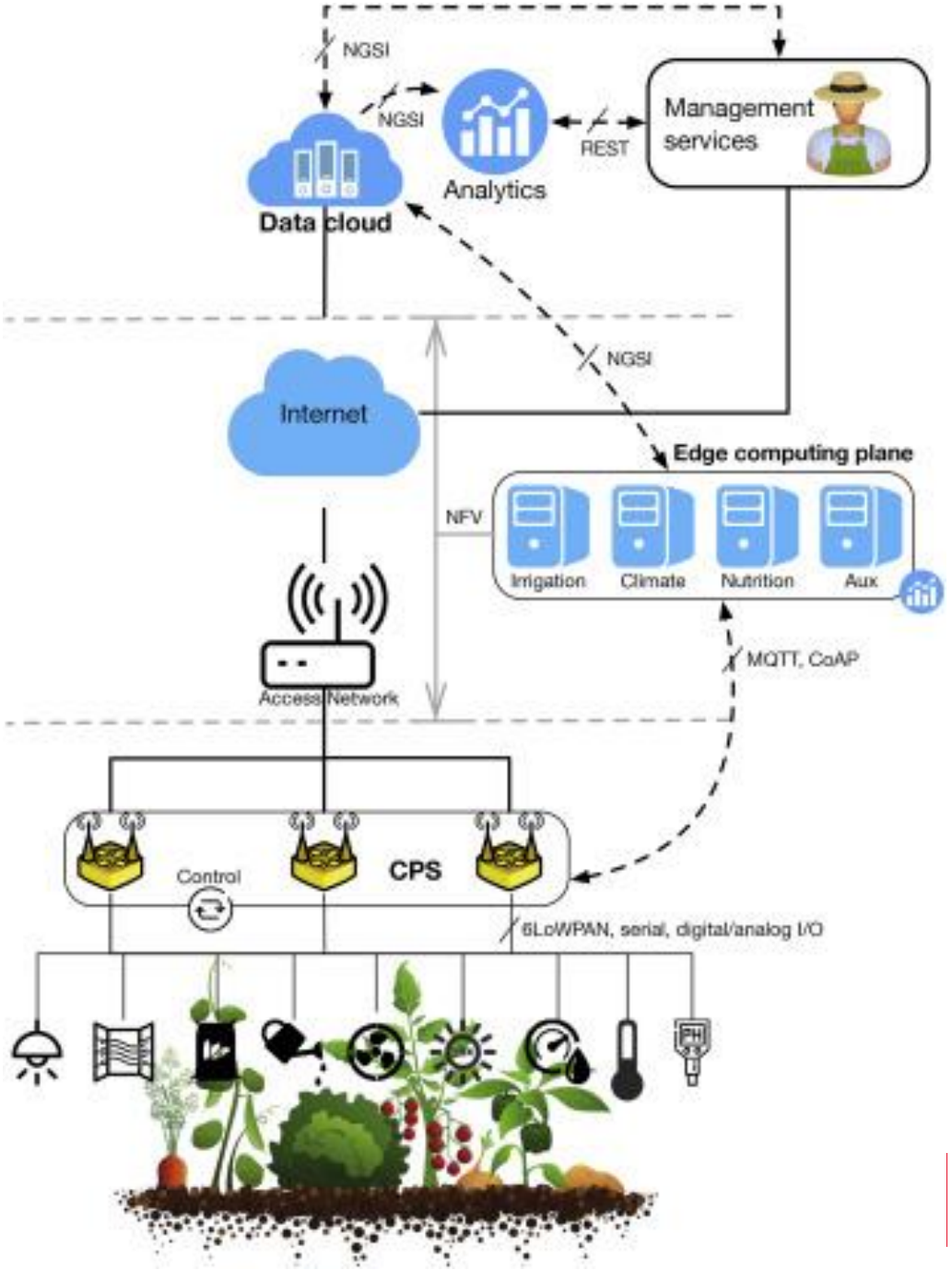


# High-Level Architecture (IoT Examples)

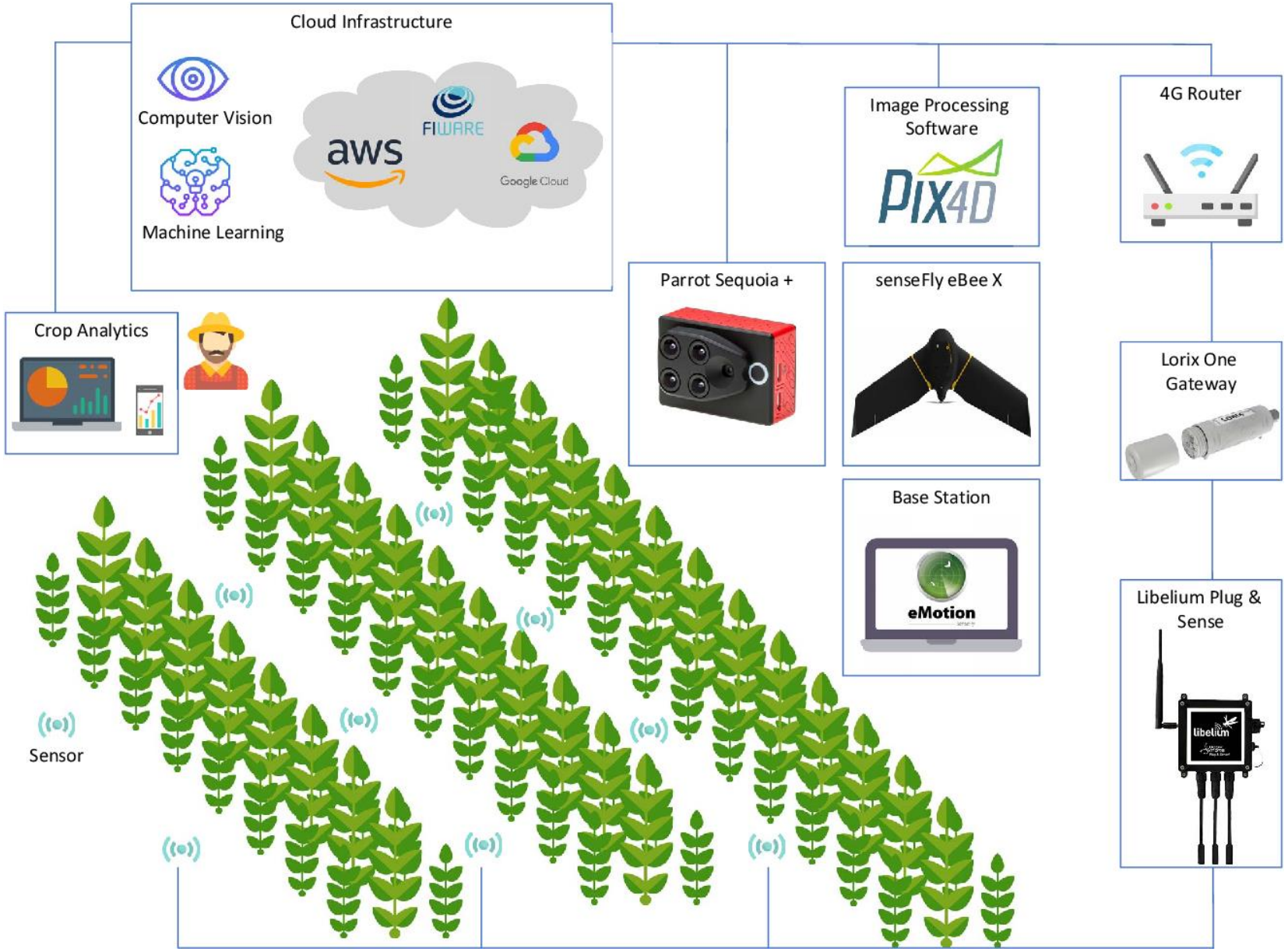


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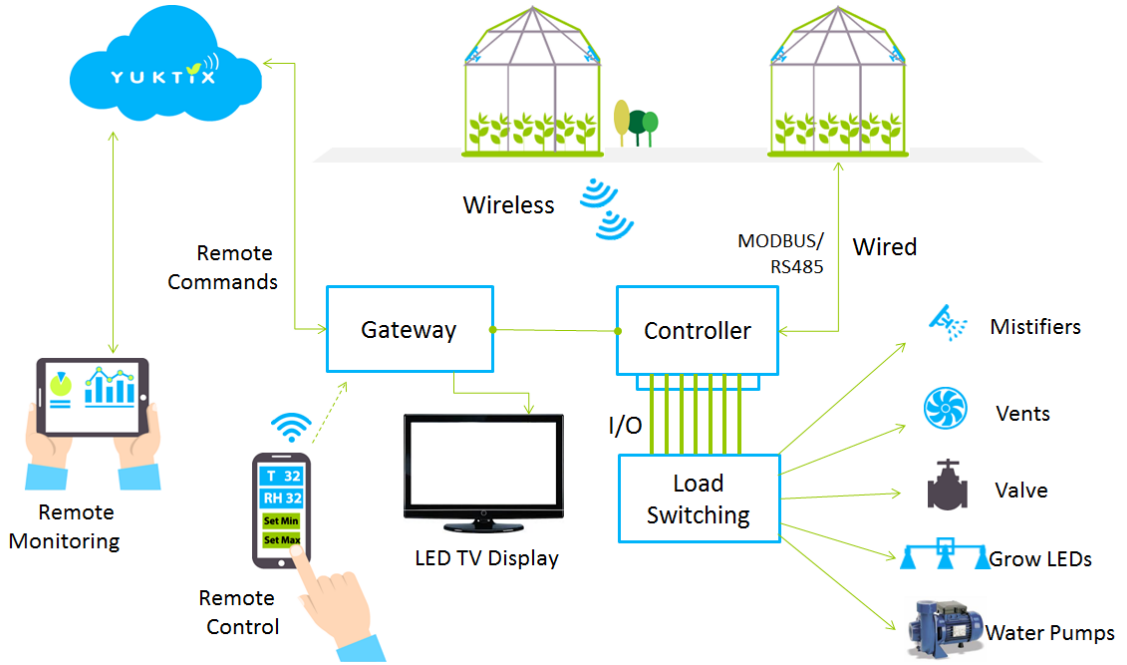
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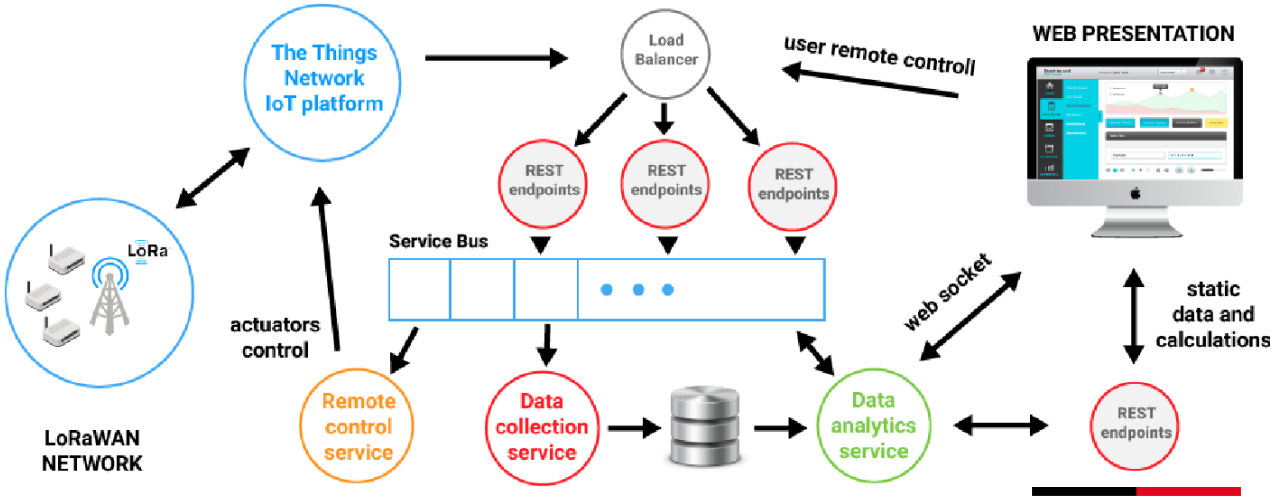


# High-Level Architecture for IoT

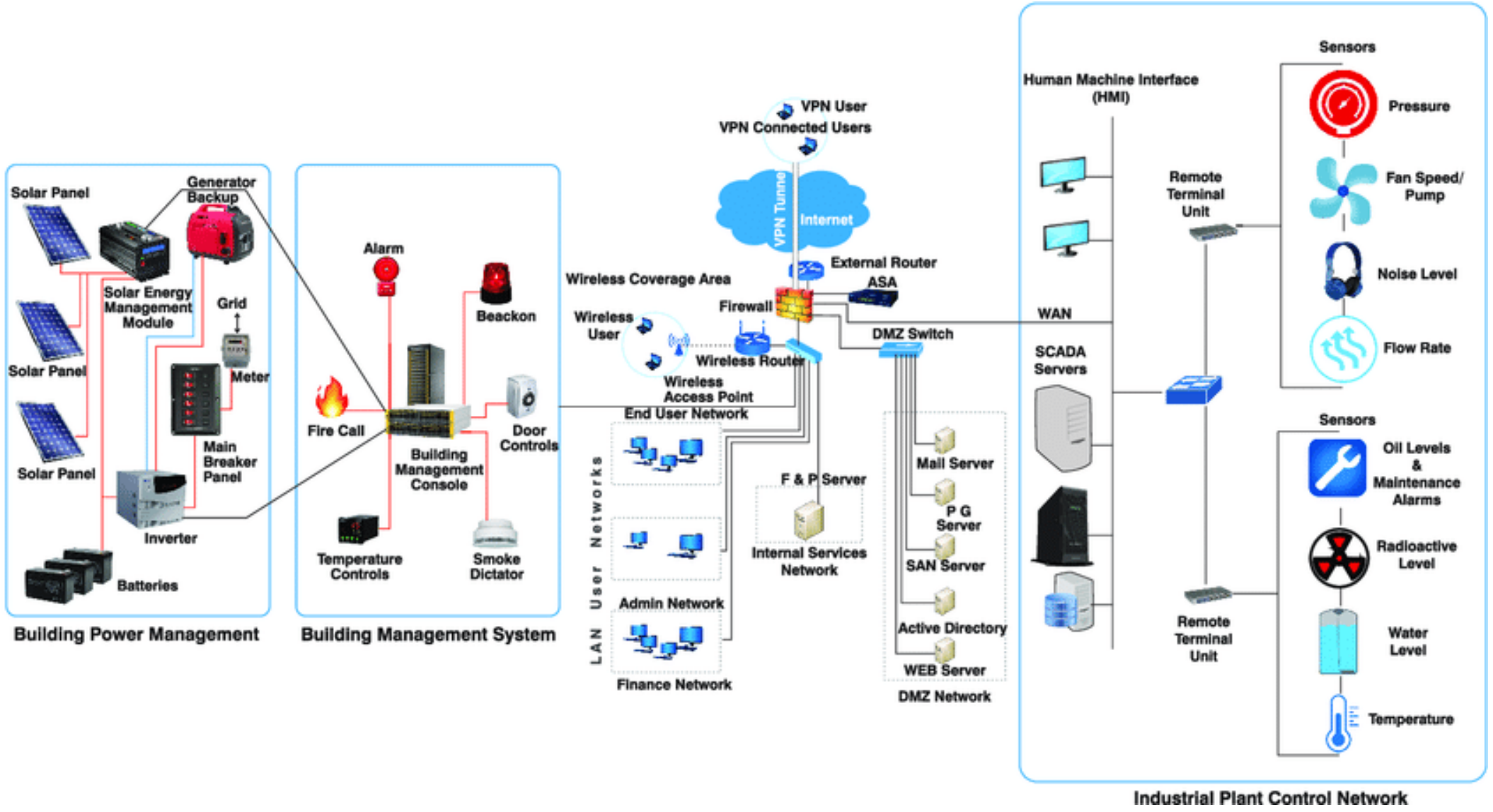
- Notice that there are no set standard for describing the High-Level Architecture Design for IoT system, because IoT systems is a Multidisciplinary (involves a combination of Computing, Engineering and Construction disciplines).

- Common practices:
  - Use a separate Block Diagram to represent the Hardware, Software and Network Architectures:
    - **Hardware**: Sensors, Actuators, Modules, MCU Platforms, Interfaces, Edge Devices, User Devices, etc.
    - **Software**: Communications Protocols (like MQTT, CoAP), Database, Cloud Platform, Services, etc.
    - **Network**: Networking Devices, Subnets, Gateways, End Devices, ISP, Cloud, etc.
  - Elaborate using the **"4+1" Views of Architecture**.

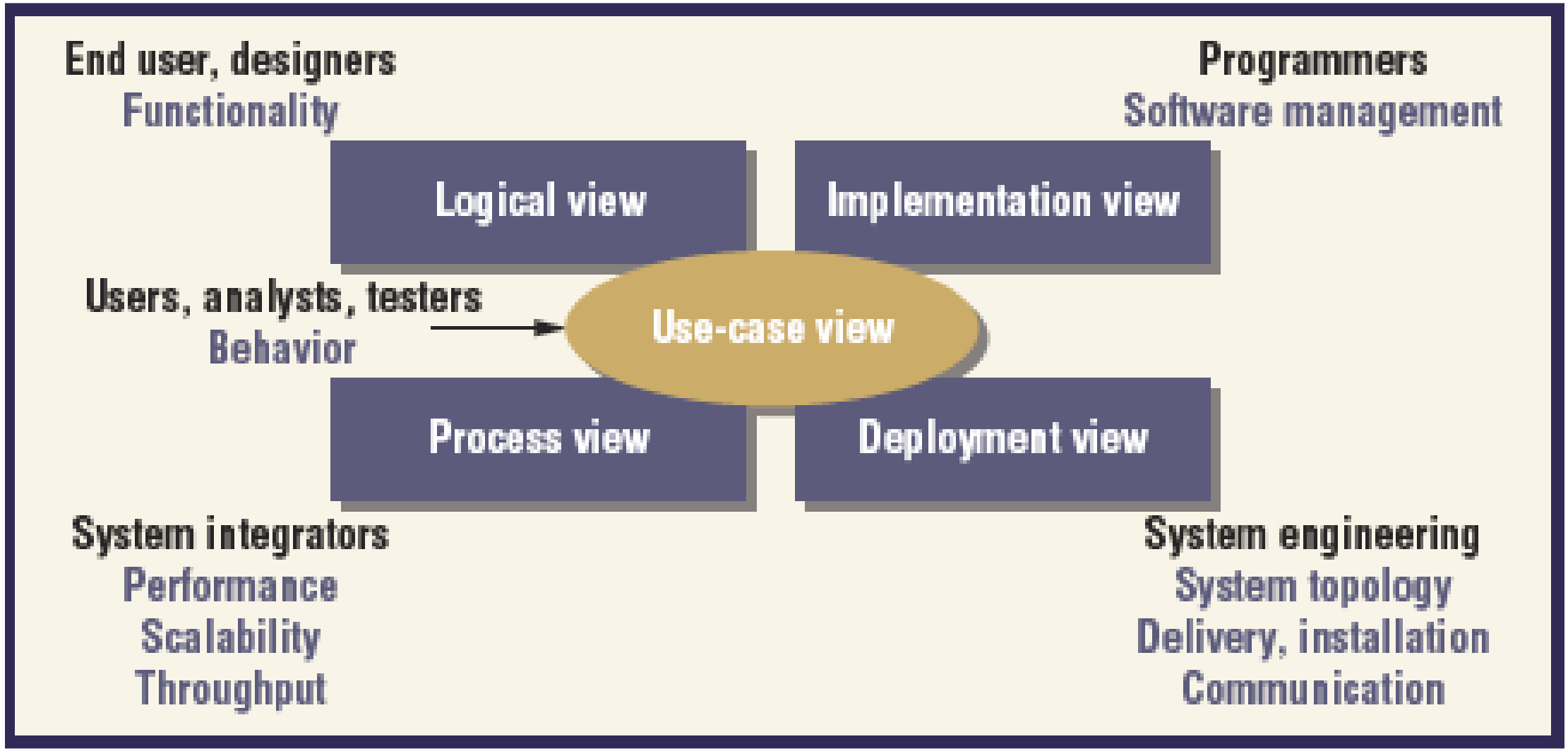


# High-Level Architecture for IoT (Cont.)



# The 4+1 Views of Architecture



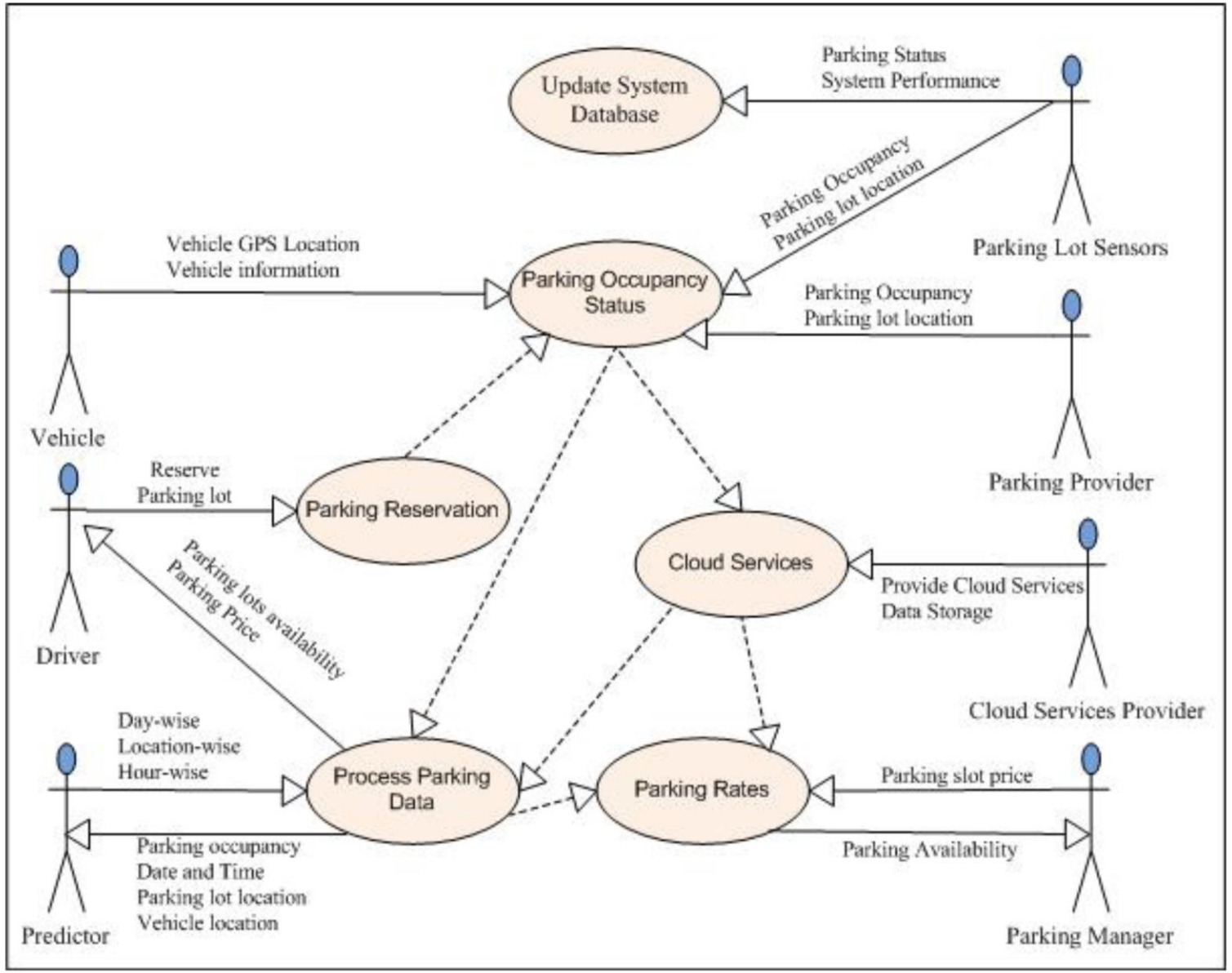

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# Use Case View

- The Use Case View shows how the Stakeholders interact with the System – translated from the Use Cases (elaborated through the Usage Scenarios).

- It also the relationship between Functions/Operations.

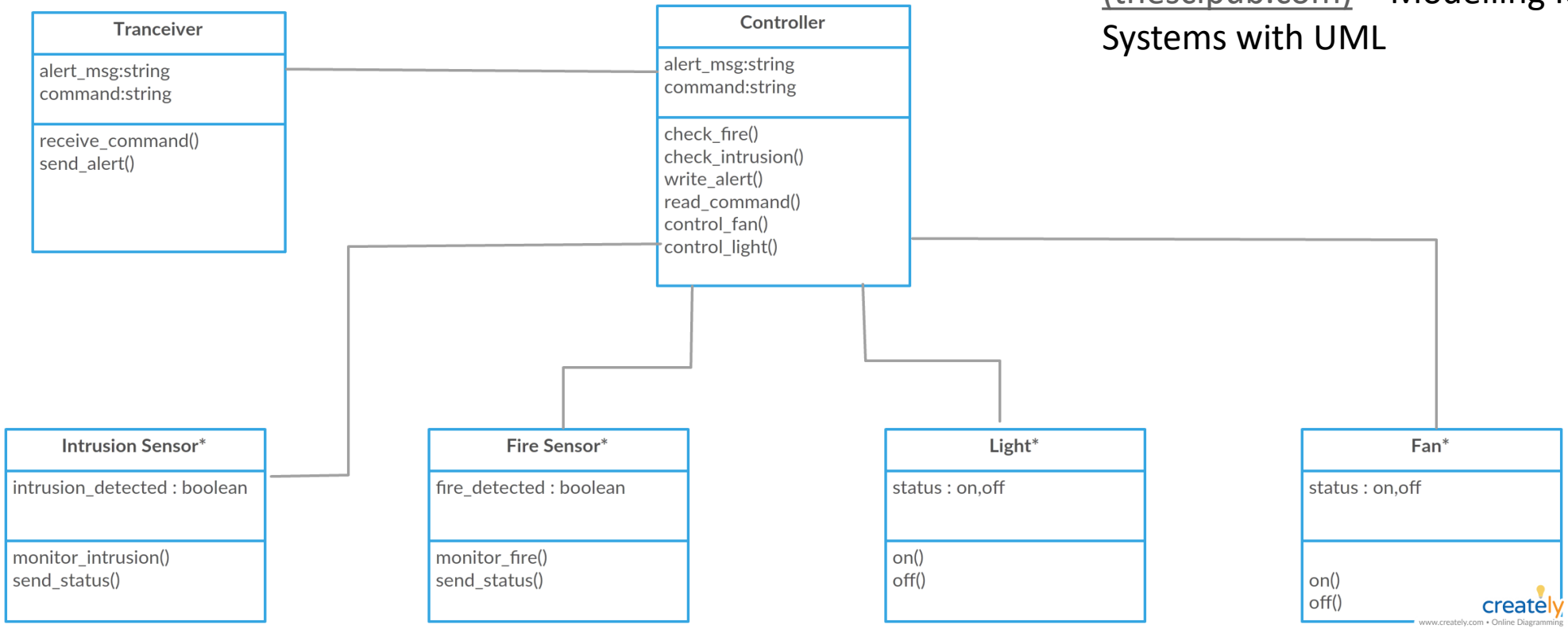
- Learn more here:
  - UML Use Case Diagram Tutorial | Lucidchart
  - What is Use Case Diagram? (visual-paradigm.com)



# Logical View

- The Logical View is a subset of the Design Model which presents architecturally significant design elements
- Describes the most important classes
- Their organization in packages and subsystems into layers
- It also describes the most important use-case realizations, for example, the dynamic aspects of the architecture

- Class Diagrams are one of the most useful ways to express the Logical Design of a system.
- Learn more about them here:
  - UML - Class Diagram (tutorialspoint.com)
  - ajeassp.2021.81.93.pdf (thescipub.com) – Modelling IoT Systems with UML

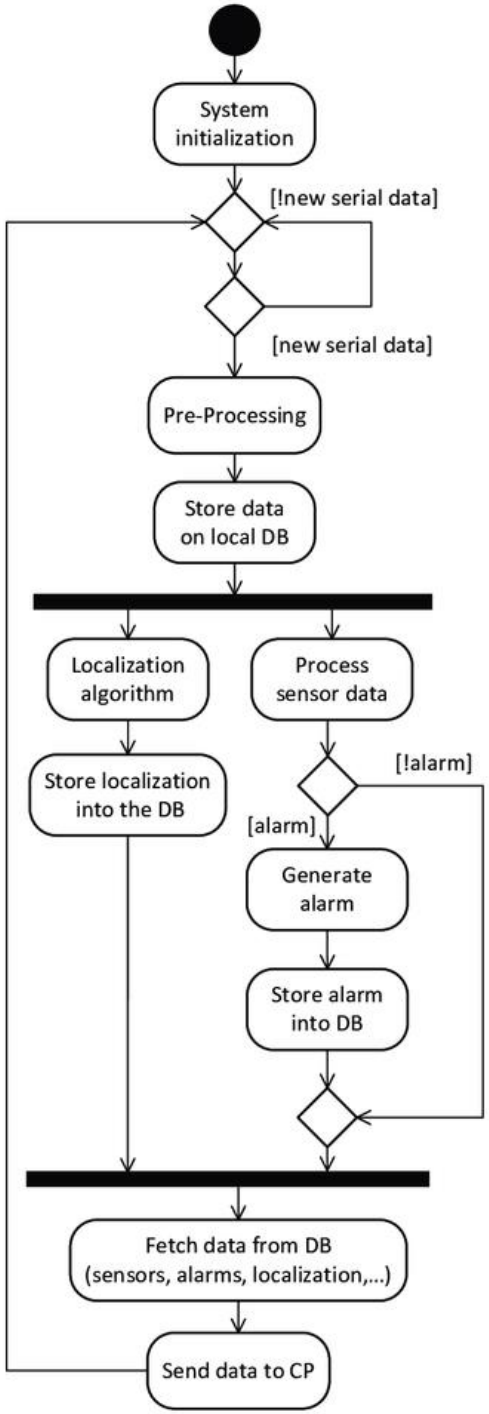


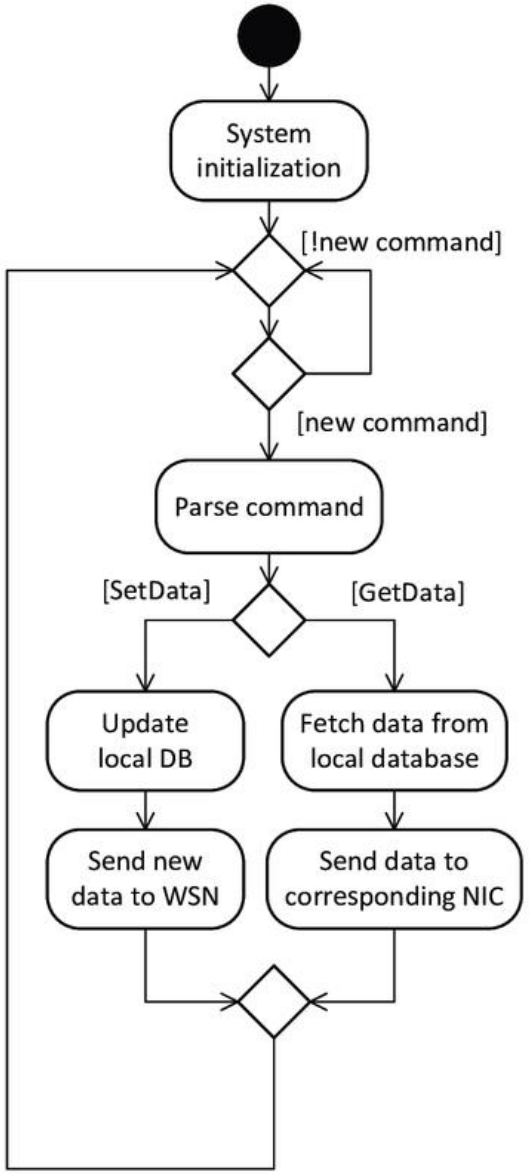

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# Process View

- Consists of the ***processes*** and ***threads*** that form the system's ***concurrency*** and ***synchronization*** mechanisms, as well as their ***interactions***
- Is captured using class, interaction and state transition diagrams with a focus on active classes and objects.

- Learn more about Activity Diagrams here:
  - UML - Activity Diagrams (tutorialspoint.com)
  - What is Activity Diagram? (visual-paradigm.com)



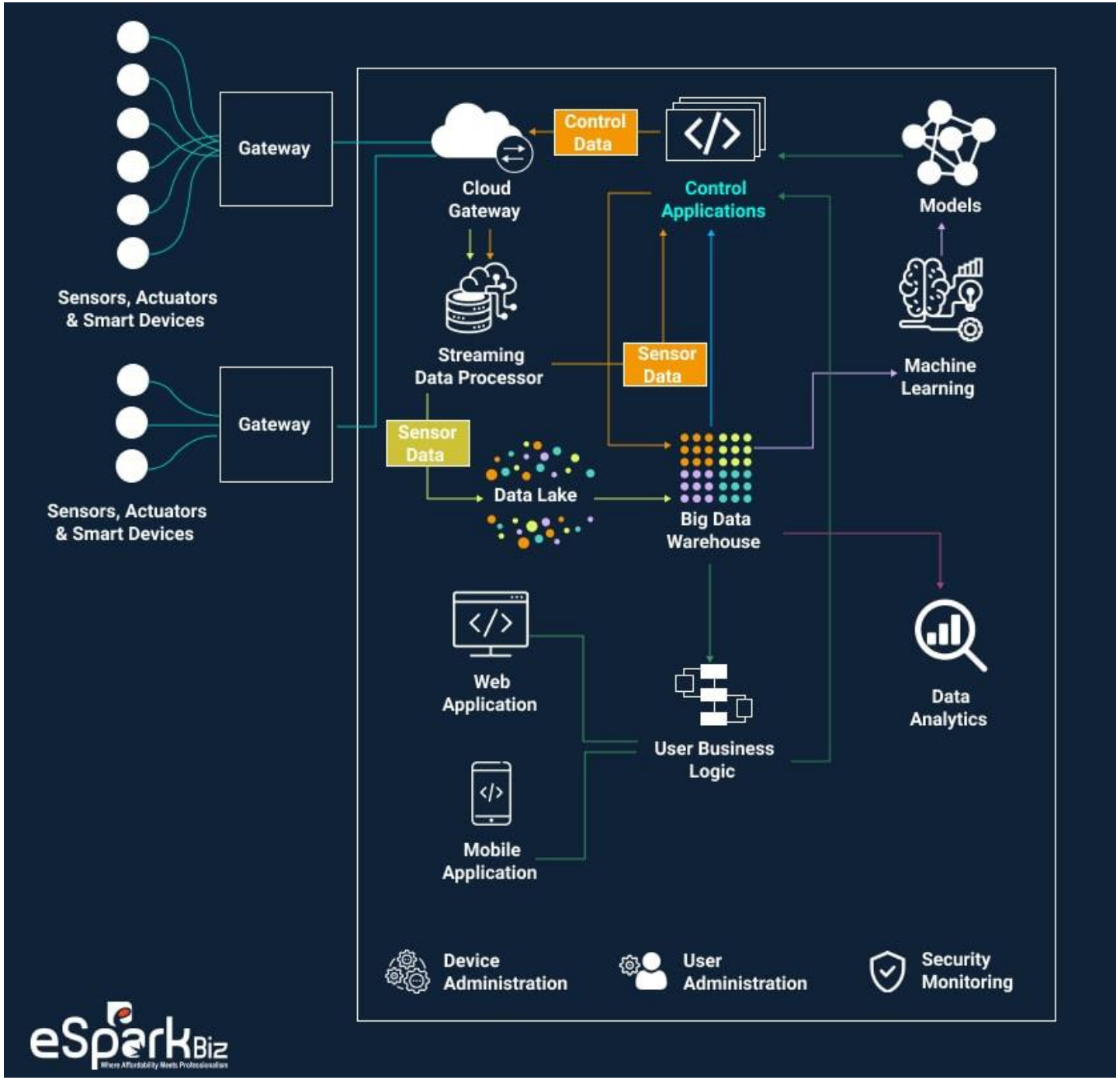



# Deployment View

- The Deployment View in Software Development is meant to describe the hardware components and networking involved in the proposed system.

- In IoT Projects, this view is represented by the 3 Block Diagrams:
  - Hardware
  - Software
  - Networking

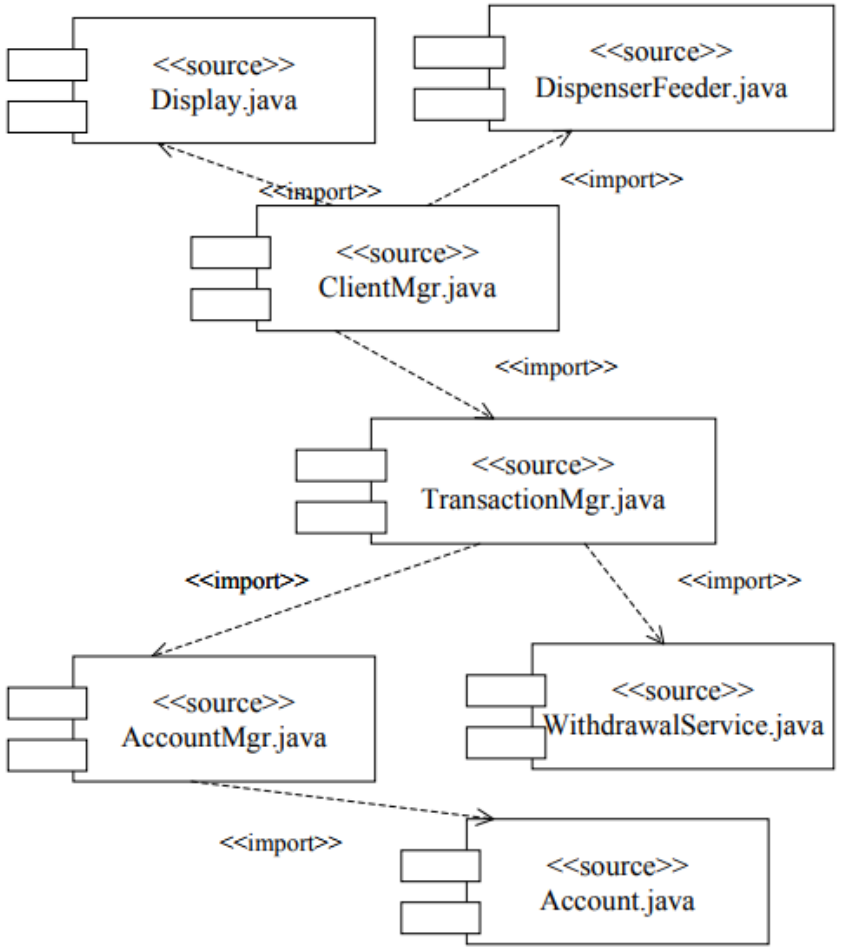
- Moving On!



# Implementation View

- Describes the **organization of static software modules** (source code, data files, executables, documentation etc.) in the development environment in terms of **Packaging and layering**
- Are modelled using UML Package Diagrams.
- UML components are physical and replaceable parts of a system that conform to and provide the realization of a set of interfaces

| Classes | Source Components |
|---|---|
| CardReader, Display, KeyPad | Display.java |
| DispenserFeeder, DispenserSensor, CashCounter | DispenserFeeder.java |
| ClientMgr | ClientMgr.java |
| TransactionMgr | TransactionMgr.java |
| AccountMgr | AccountMgr.java |
| WithdrawalServic e | WithdrawalService.java |
| Account, Persistent class | Account.java |



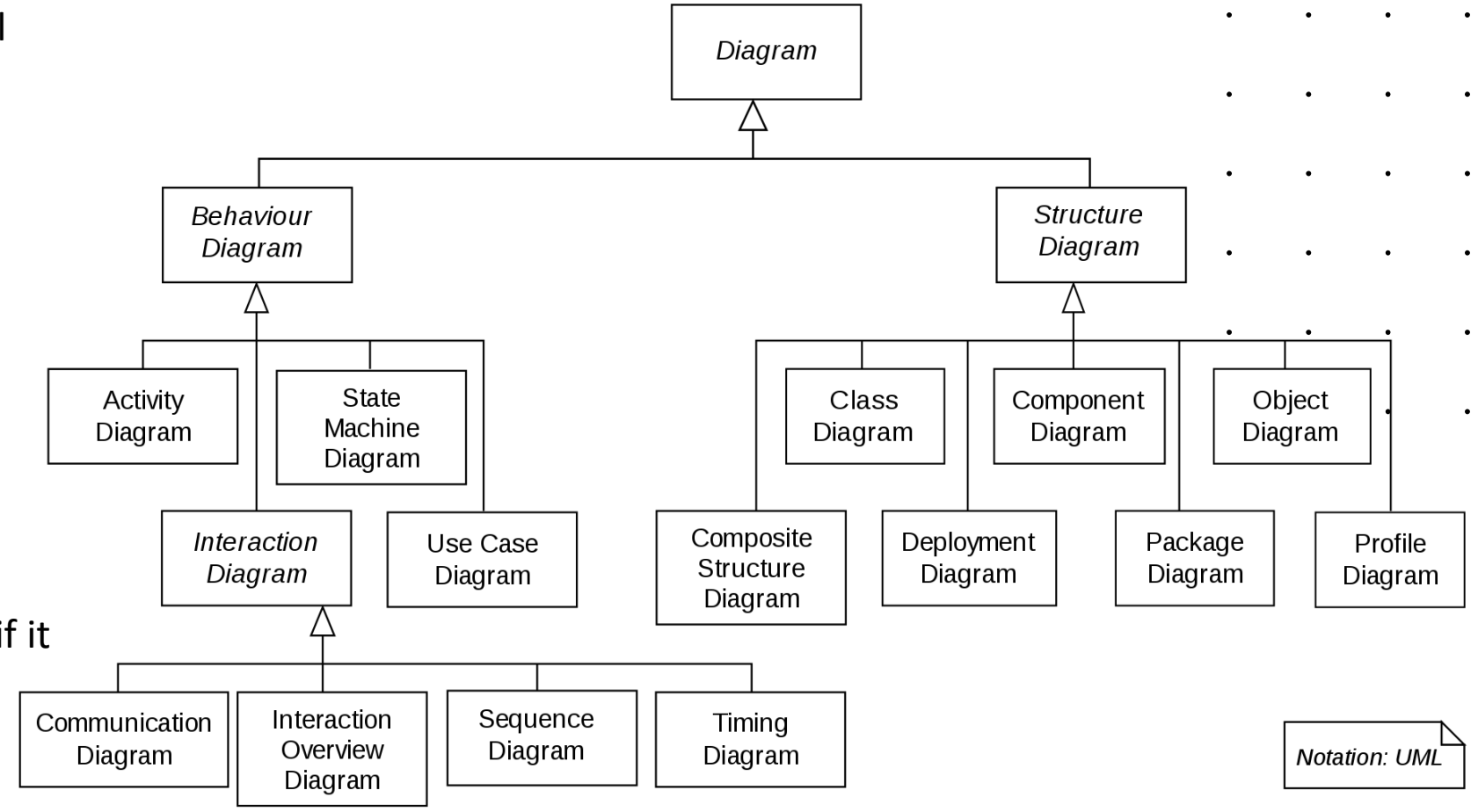

- However, in IoT Projects, the Implementation View is often represented by the Software Block Diagram.
- Moving On!

# UML

- The Unified Modelling Language is a standard set of diagrams that all software engineers can use for building and understanding Architecture Designs.

- Learn about them here:
  - What is Unified Modeling Language (UML)? (visual-paradigm.com)
  - UML Tutorial (tutorialspoint.com)

- You can use diagramming tools if it supports UML 2.x, such as:
  - Visual Paradigm
  - Draw.IO
  - StarUML
  - Microsoft Visio
  - LucidChart



# Takeaway for this Week

1. Follow the steps in IoT Project Proposal process to identify your “Scope” of IoT work.

2. Your IoT System Design consists of 2 stages of Diagrams:
   A. High-Level Architecture
      - Usually in 3 diagrams: Hardware, Software and Network Architecture

   B. Low-Level Architecture
      - Circuit Diagram
      - Flowcharts
      - Use Case, Class, ERD, or etc. diagrams needed for software design

3. Pick one of the tools available and look for online tutorials on how to use them to draw all the diagrams you need.

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**Presented by Dr. Mark Tee Kit Tsun (SUTS)**

If you have questions, find me on MS Teams: mtktsun@swinburne.edu.my